Firm Dynamics and Directed Technical Change under Liquidity Constraints

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October 23, 2026

Internal Macroeconomics Seminar

Ghent, Belgium

Today's Menu

- Introduction
- Stylized Example (← Today's focus!)
- General Model
- To Be Done: Data, Identification and Estimation
- Research 2

Motivation: Background

- **Climate policies** have been at the center stage of recent policy debates.
- Governments want to achieve **environmental goals** while maintaining **economic growth**.

- The EU Green Deal views **environmental policy as a new growth strategy**.
- EU countries have subsidized **energy-saving** and **non-energy-input-saving** innovation.

- **Q. Is it possible to achieve both environmental goals and economic growth?**

Motivation: Empirical Regularities

- Not all firms engage in research and development (R&D):
  e.g., In Belgium, energy-saving (58%) vs. non-energy-input-saving (39%)

- Innovating firms are subject to **liquidity constraints**:
  i.e., Firms have to borrow some money before starting innovation.

- This paper studies firms' directed technical change under **financial frictions**:
  - My framework tracks **environmental goals** and **economic growth**.
  - My framework captures **heterogeneity in firms' self-selection into R&D investments**.

▸ Literature

Setup

- A simple closed-economy, single-period model:
  - No uncertainty
  - Partial equlibrium

- Three agents: Household, Firms, and Government:
  - Government provides firms with innovation subsidies.
  - Household's utility is a constant-elasticity-substitution (CES) function of firms' products.
  - A continuum of firms are monopolistic.

Firm's Problem: Production Function

- The firm's problem proceeds in two steps:
  1. R&D investment decision (subject to liquidity constraints)
  2. Production decision

- Firm i's output q_i is a Cobb-Douglas composite of non-energy input x_i and energy e_i:

$$q_i = (z_i^x x_i)^\beta (z_i^e e_i)^{1-\beta},$$

  - z_i^x and z_i^e: input-saving and energy-saving technologies
  - β: the output elasticity of x_i

Firm's Problem: Productivity Process

- Productivity z_i^e is a function of R&D input (e.g., scientists) s_i^e:

$$z_i^e = (1 + \gamma s_i^e)^\delta \bar{z}_i^e.$$

where $\bar{z}_i^e$ is the status-quo level of productivity, and $\gamma, \delta \in (0, 1)$ are parameters.

$\hookrightarrow$ The marginal productivity at $s_i^e = 0$ is $\delta \bar{z}_i^e < \infty$.

- In the existing literature,

$$z_i^e = (1 + \gamma s_i^{e\delta}) \bar{z}_i^e.$$

$\hookrightarrow$ The marginal productivity at $s_i^e = 0$ is ∞.

- z_i^x is analogous.

Firm's Problem: Production and Innovation Investment

- Given s_i^x and s_i^e, the firm maximizes its profit while minimizing its input costs.
  $\Longrightarrow$ the production profit $\pi(s_i^x, s_i^e)$
- The firm chooses s_i^x and s_i^e to maximize its profit net of R&D investment.
- The firm is faced with liquidity constraints:

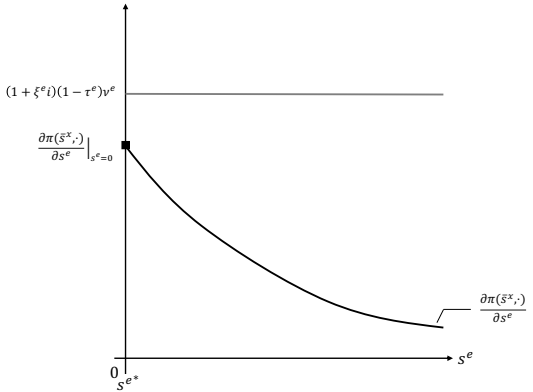
$$\left.\begin{array}{l} \text{- the cost of R\&D investment: } \nu^e s_i^e \\ \text{- the cost of borrowing: } i\xi_i^e \nu^e s_i^e \\ \text{- the ad valorem innovation subsidy: } 1-\tau^e \end{array}\right\} (1-\tau^e)(1+i\xi_i^e)\nu^e s_i^e$$

  where ν^e: wage for scientists, ξ_i^e: tightness of the firm's financial constraint, and i: return rate of borrowing.

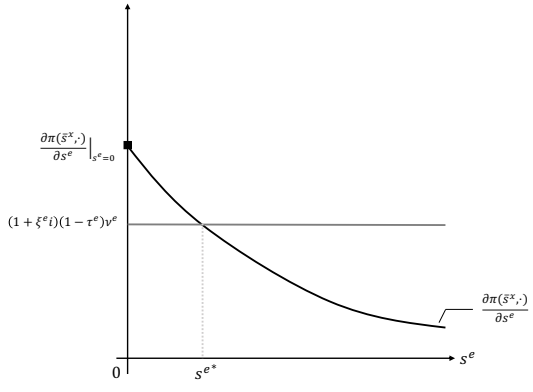
- s_i^x is analogous.

Firm's Problem: Innovation Investment

- The firm chooses s_i^e to maximize $\pi(\bar{s}_i^x, s_i^e) - (1-\tau^e)(1+i\xi_i^e)\nu^e s_i^e$.
- There can be two scenarios:



(A) No R&D investment



(B) Positive R&D investment

Conventional Approach

Firm's Problem: Innovation Investment

- The choice of s_i^e depends on the choice of s_i^x, and vice versa.

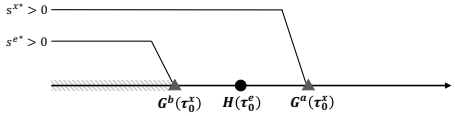
- The condition for interior solutions can be expressed in terms of τ and τ^e:

$$s_i^{x*} > 0 \iff H_i(\tau^e) < G_i^a(\tau^x) \qquad \text{and} \qquad s_i^{e*} > 0 \iff H_i(\tau^e) < G_i^b(\tau^x),$$

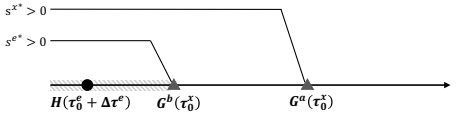
  where $G_i^a(\cdot)$ and $G_i^b(\cdot)$ are **monotonically increasing** in τ^x, whereas $H_i(\cdot)$ is **monotonically decreasing** in τ^e.

Firm's Problem: Innovation Investment

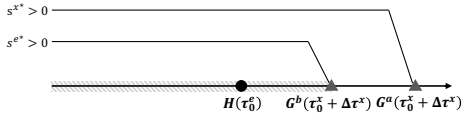
- Suppose the current subsidies τ_0^x and τ_0^e are such that $G_i^b(\tau_0^x) < H_i(\tau_0^e) < G_i^a(\tau_0^x)$: i.e., $s_i^x > 0$ and $s_i^e = 0$
- The government wants the firm to engage in **energy-saving innovation** (i.e., $s_i^e > 0$).
- This can be achieved by increasing innovation subsidies ($\Delta\tau^x, \Delta\tau^e > 0$):



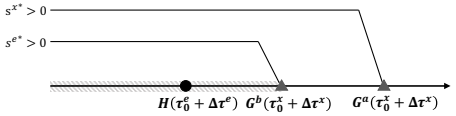
(A) Status-quo policy regime



(B) $\Delta\tau^e$ only



(C) $\Delta\tau^x$ only



(D) Both $\Delta\tau^x$ and $\Delta\tau^e$

Three Takeaways

- The firm's response is determined by $G_i^a(\cdot)$, $G_i^b(\cdot)$, and $H_i(\cdot)$.

1 Firms' responses are heterogeneous.
$\hookrightarrow$ The policy analysis needs to keep track of firm-level responses.

2 Market distortions arise at the firm level, while policies work at the aggregate level.
$\hookrightarrow$ The policy analysis compares outcome variables across different policy regimes.

3 Firms' responses depend on "deep parameters" and aggregate variables.
$\hookrightarrow$ The policy analysis should be nonparametric and general-equilibrium.

Firm-Level Production Function

- Firm's production function:

$$q_i = f(f^x(\ell_i, k_i; z_i^x), f^e(e_i; z_i^e)),$$

  where ℓ_i: labor, and k_i: capital.

- Productivity process:

$$z_i^x = h^x(s_i^x; \bar{z}_i^x).$$

- $h^x(\cdot)$ is continuously differentiable with bounded derivatives.
  $\Longrightarrow$ The marginal productivity at $s_i^x = 0$ is finite! (z_i^e is analogous.)

- The production and innovation decisions are exactly the same as before.

Object of Interest

- Consider a policy reform from $\boldsymbol{\tau}^0$ to $\boldsymbol{\tau}^1$.
- The changes in energy use and GDP:

$$\Delta E(\boldsymbol{\tau}^0, \boldsymbol{\tau}^1) := E(\boldsymbol{\tau}^1) - E(\boldsymbol{\tau}^0) \qquad \text{and} \qquad \Delta Y(\boldsymbol{\tau}^0, \boldsymbol{\tau}^1) := Y(\boldsymbol{\tau}^1) - Y(\boldsymbol{\tau}^0),$$

  where $E(\boldsymbol{\tau})$ is aggregate energy use under the policy $\boldsymbol{\tau}$; and $Y(\boldsymbol{\tau})$ is GDP under $\boldsymbol{\tau}$.

- EU Green Deal: "**The environmental policy is a new growth strategy.**"
  $\hookrightarrow$ **Q. Can it be true that $\Delta E(\boldsymbol{\tau}^0, \boldsymbol{\tau}^1) < 0$ and $\Delta Y(\boldsymbol{\tau}^0, \boldsymbol{\tau}^1) > 0$?**

Data

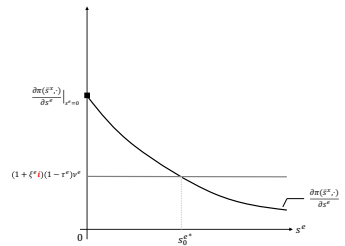
- Eurostat:
  ↪ aggregate variables (e.g., wage, price index, GDP)
- Orbis Europe:
  ↪ firm-level labor, capital, and revenue.
- Stabel:
  ↪ firm-level energy use
- CIS (BELSPO) and EIBIS:
  ↪ firm-level R&D investment

Identification and Estimation

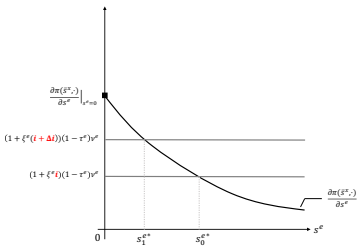
- I will develop a new identification procedure for the firm's production function.
  - Identification from revenue data (Chow, Kasahara and Sugita 2026)
  - Identification in the presence of R&D investment (Doraszelski and Jaumandreu 2013)
- My approach blends these papers.

Research 2

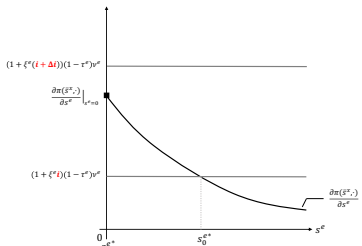
- Environmental policies (possibly) induce **inflation**.
- The monetary authority should raise the interest rate: $\Delta i > 0$.
- The monetary policy may have implications for firms' R&D investment:



(A) Status-quo policy regime



(B) Small Δi



(C) Large Δi

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Motivation: Empirical Regularities

Empirical literature on liquidity constraints:

- Brown, Martinsson and Petersen (2012); Hottenrott and Peters (2012); Gorodnichenko and Schnitzer (2013):

  $\hookrightarrow$ the limited availability of internal sources and the restricted access to external sources for R&D investments

  $\hookrightarrow$ EU firms

- Brown, Fazzari and Petersen (2009)

  $\hookrightarrow$ U.S. firms

back

Motivation: Empirical Regularities

Endogenous growth models:

- Klette and Kortum (2004); Lentz and Mortensen (2008); Akcigit and Kerr (2018):
  ↪ Schumpertarian growth models
  ↪ All firms invest in R&D
- Acemoglu, Akcigit, Alp, Bloom and Kerr (2018):
  ↪ Schumpertarian growth models
  ↪ Firms can choose not to enter
- Atkeson and Burstein (2019)
  ↪ A variety-expansion model
- Buera, Kaboski and Shin (2011); Moll (2014); Midrigan and Xu (2014):
  ↪ capital investment under financial frictions
- Capelle, Diaz, Kirti, Villegas-Bauer and Banerjee (2025); Ottonello and Winberry (2026):
  ↪ allocation of resources between production capital and innovation input.

back

Motivation: Empirical Regularities

Directed Technical Change:

- Acemoglu, Aghion, Bursztyn and Hemous (2012); Acemoglu, Akcigit, Hanley and Kerr (2016); Aghion, Dechezleprêtre, Hémous, Martin and Van Reenen (2016); Fried (2018); Aghion, Barrage, Donald, Hémous and Liu (2025):

  $\hookrightarrow$ Clean vs. dirty technologies

- Hassler, Krusell and Olovsson (2015, 2021); Farajpour and Hinkelmann (2021); Casey (2024); Hinkelmann (2024a,b,c); Känzig and Williamson (2025); Çakır-Melek and Orak (2026)

  $\hookrightarrow$ Energy-saving vs. non-energy-input-saving technologies

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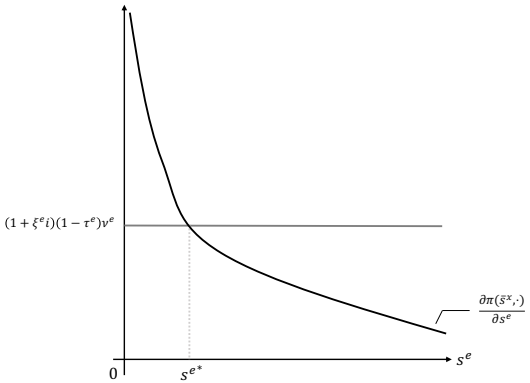
Motivation: Empirical Regularities

Directed Technical Change (cont'd):

- Aghion, Bergeaud, Ridder and Reenen (2024):
  - ↪ Clean vs. dirty technologies under liquidity constraints
  - ↪ All firms invest in R&D
- Fornaro, Guerrieri, Hotten and Reichlin (2026):
  - ↪ Clean vs. dirty technologies under liquidity constraints
  - ↪ Empirical research

back

Conventional Approach



(C) Positive R&D investment

back

R&D Switching

- Suppose: the current policy regime $\{\tau_0^x, \tau_0^e\}$ is such that $G_i^b(\tau_0^x) < H_i(\tau_0^e) < G_i^a(\tau_0^x)$.
- Assume: $G_i^a(\cdot)$, $G_i^b(\cdot)$, and $H_i(\cdot)$ are continuously differentiable in their arguments.
- Consider changing both input- and energy-saving innovation subsidies by $\Delta\tau^x$ and $\Delta\tau^e$, respectively.
- Firm i switches from input-saving to energy-saving R&D activity if and only if

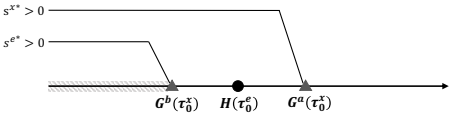
$$H_i(\tau_0^e) - G_i^a(\tau_0^x) \geq -\int_{\tau_0^e}^{\tau_0^e+\Delta\tau^e} \frac{\partial H_i(\cdot)}{\partial \tau^e} d\tau^e + \int_{\tau_0^x}^{\tau_0^x+\Delta\tau^x} \frac{\partial G_i^a(\cdot)}{\partial \tau^x} d\tau^x, \tag{1}$$

and

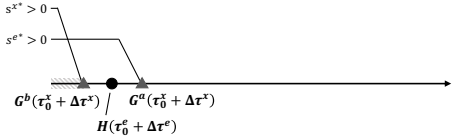
$$H_i(\tau_0^e) - G_i^b(\tau_0^x) \leq -\int_{\tau_0^e}^{\tau_0^e+\Delta\tau^e} \frac{\partial H_i(\cdot)}{\partial \tau^e} d\tau^e + \int_{\tau_0}^{\tau_0^x+\Delta\tau^x} \frac{\partial G_i^b(\cdot)}{\partial \tau^x} d\tau^x. \tag{2}$$

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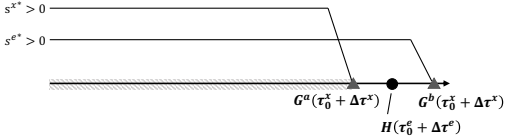
R&D Switching



(A) Status-quo policy regime



(B) $\Delta\tau^x < 0$ and $\Delta\tau^e > 0$



(C) $\Delta\tau^x > 0$ and $\Delta\tau^e < 0$

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